23. Number Codes (Set B)

Year 4 · Mark Scheme — Supplementary Worksheet

Foundation — Answers

1. BRD

2. 416

3. A — ATM

4. JOY

5. 056

6. CAT

7. 139

8. EON

9. 123

10. DOG

11. 368

12. GEL

13. 409

14. A and D (612 and 673 both begin with the digit 6, which codes as T; 348 begins with 3=A, 489 begins with 4=R, and 256 begins with 2=D)

15. 567

Standard — Answers

16. 15 (6×2=12, then 12+3=15)

17. 11 ((18+4)÷2=11)

18. C — 13 (9×2=18, then 18−5=13)

19. 22 ((8+3)×2=22)

20. 12 ((10−4)×2=12)

21. B — 7 ((25−4)÷3=7)

22. 25 (the rule is ×4, then +1: 2×4+1=9, 4×4+1=17, so 6×4+1=25)

23. 4 (the rule is ×4, then +1: 3×4+1=13, 6×4+1=25, so 17=4×x+1 means x=4)

24. D and E (5×4+2=22, 6×4+2=26, 7×4+2=30, 8×4+2=34, 9×4+2=38; only 8 and 9 give a result greater than 30 — 7 gives exactly 30, which is not greater)

25. C — 43 (the rule is ×4, then +3: 4×4+3=19, 7×4+3=31, so 10×4+3=43)

26. 20 ((7−2)×4=20)

27. 6 ((38−2)÷6=6)

28. C — 25 ((8−3)×5=25)

29. 28 (the rule is ×3, then +1: 3×3+1=10, 5×3+1=16, so 9×3+1=28)

30. 7 (the rule is ×3, then −1: 4×3−1=11, 9×3−1=26, so 20=3×x−1 means x=7)

31. 29 (11×2=22, then 22+7=29)

32. 10 ((14+6)÷2=10)

33. B — 24 ((7+5)×2=24)

34. 12 ((15−9)×2=12)

35. 36 (the rule is ×4, then +4: 2×4+4=12, 5×4+4=24, so 8×4+4=36)

Hard — Answers

36. 617 (2→6, 7→1 wraps round, 3→7)

37. 371 (subtract 4 from each digit, wrapping: 7→3, 1→7 wraps round, 5→1)

38. A — 941 (3→9, 8→4 wraps round, 5→1)

39. 926 (629 reversed is 926)

40. A, C and D (157 has a 5, which becomes 0, giving 602; 358 has a 5, giving 803; 503 has a 5, giving 058; 246 and 629 contain no 5, so their codes have no 0 digit)

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